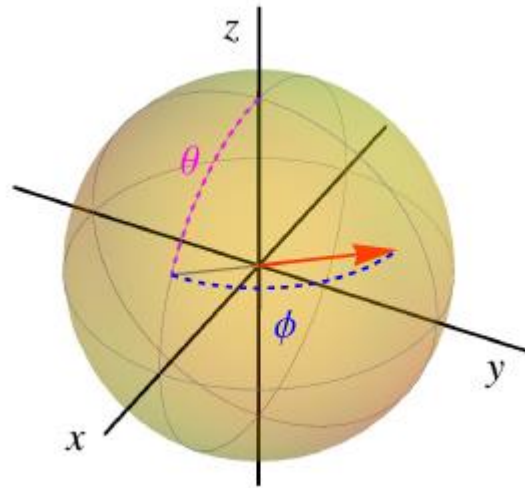


Quantum Computation

ELK336E

Week #9



Density operator

Density operator is another formulation to describe the state of a quantum system.

The first postulate of a quantum system asserts that a state vector $|\psi\rangle$ has the full information about a quantum system.

This state is called a **pure state**.

Now, consider a collection (an ensemble) of pure states.

The quantum state of this ensemble is described by a density operator, which is in general in a mixed state.

$$\rho \equiv \sum_i p_i |\psi_i\rangle \langle \psi_i|. \quad (56)$$

A pure state $|\psi\rangle$ can also be represented by a density matrix as follows;

$$\rho = |\psi\rangle \langle \psi| \quad (57)$$

Consider, $|\psi\rangle = a|0\rangle + b|1\rangle$; corresponding density operator is ;

$$\rho = (a|0\rangle + b|1\rangle)(a^*\langle 0| + b^*\langle 1|)$$

$$\rho = aa^*|0\rangle\langle 0| + ab^*|0\rangle\langle 1| + ba^*|1\rangle\langle 0| + bb^*|1\rangle\langle 1|$$

Here we can write ;

$$aa^* = |a|^2, bb^* = |b|^2, ab^* = c, b^*a = c^*$$

$$\rho = \begin{bmatrix} |a|^2 & c \\ c^* & |b|^2 \end{bmatrix} \quad (58)$$

$|a|^2$ probability that the system is in the $|0\rangle$ state.

$|b|^2$ probability that the system is in the $|1\rangle$ state.

Therefore
 $|a|^2 + |b|^2 = 1.$

c : coherence parameter
 (the presence of this parameter indicated that $|0\rangle$ and $|1\rangle$ are in a coherent superposition state.)

PROPERTIES OF A DENSITY OPERATOR

An operator ρ is a density operator if and only if it satisfies the following three requirements:

- The density operator is Hermitian, meaning $\rho = \rho^\dagger$.
- $Tr(\rho) = 1$.
- ρ is a positive operator, meaning $\langle u|\rho|u\rangle \geq 0$ for any state vector $|u\rangle$.

Recall that an operator is positive if and only if it is Hermitian and has nonnegative eigenvalues.

Additionally a density operator represents a pure state if its purity factor is $\wp = 1$, where

$$\wp = \text{Tr}(\rho^2).$$

In general purity is between

$\frac{1}{d} \leq \wp \leq 1$ where d is the dimension of the Hilbert space representing the system.

- $\text{Tr}(\rho^2) < 1$ for a mixed state.
- $\text{Tr}(\rho^2) = 1$ for a pure state.

Expectation Values

The expectation value of an operator A with respect to a density operator is

$$\langle A \rangle = \text{Tr}(\rho A) \quad (59)$$

Measurement Result

In terms of the more general measurement operator formalism, the probability of obtaining measurement result m associated with measurement operator M_m is

$$P(m) = \text{Tr}(M_m^\dagger M_m \rho) \quad (60)$$

After a measurement described by a projection operator, the system is in the state described by

$$\rho \rightarrow \frac{P_n \rho P_n}{\text{Tr}(P_n \rho)} \quad (61)$$

Example Does the matrix

$$\rho = \begin{pmatrix} \frac{1}{4} & \frac{1-i}{4} \\ \frac{1-i}{4} & \frac{3}{4} \end{pmatrix}$$

represent a density operator?

We see that the matrix has unit trace

$$\text{Tr}(\rho) = \frac{1}{4} + \frac{3}{4} = 1$$

However, Since the matrix is not Hermitian, it cannot represent a density operator.

$$\rho^\dagger = \begin{pmatrix} \frac{1}{4} & \frac{1+i}{4} \\ \frac{1+i}{4} & \frac{3}{4} \end{pmatrix} \neq \rho$$

Time Evolution of the Density Operator

Recall that

$$i\hbar \frac{d}{dt} |\psi\rangle = H |\psi\rangle \qquad |\psi(t)\rangle = U |\psi(0)\rangle \qquad (62)$$

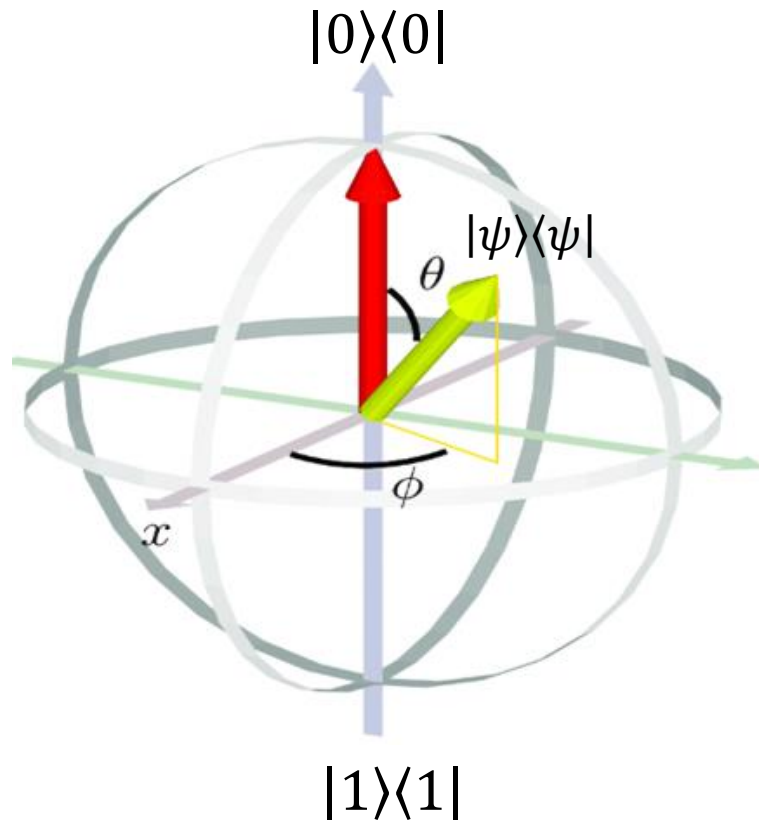
The differential equation describing the time development of a density operator can be defined by Quantum Liouville equation.

$$i\hbar \frac{d\rho}{dt} = [H, \rho] \qquad (63)$$

For a closed system the time evolution of the density operator can also be written in terms of a unitary operator U .

$$\rho(t) = U\rho(t_0)U^\dagger$$

where $U = e^{-iHt}$.



$$\rho = \frac{I + \vec{r} \cdot \vec{\sigma}}{2}$$

Here,

$$\vec{\sigma} = \begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{bmatrix}, \text{ for a pure state,}$$

$$\|\vec{r}\| = 1 \text{ and}$$

$$\vec{r} = \begin{bmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{bmatrix}.$$

$$\rho = \begin{bmatrix} \frac{1 + \cos\theta}{2} & \frac{e^{i\phi}}{2} \sin\theta \\ \frac{e^{-i\phi}}{2} \sin\theta & \frac{1 - \cos\theta}{2} \end{bmatrix}$$

NOTE: If $\vec{r} \cdot \vec{\sigma} = 0$

$$\rho = \frac{I}{2}.$$

$$\rho = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$

This is called maximally mixed quantum state.

Density matrix versus state vector

	Density matrix	State vector
Representation	$\rho = \sum_i p_i \psi_i\rangle\langle\psi_i $	$ \psi\rangle$
Pure States	YES	YES
Mixed states	YES	NO
Time evolution	$U\rho U^\dagger$	$U \psi\rangle$
Expectation value	$\text{Tr}[\rho A]$	$\langle\psi A \psi\rangle$
Multi-partite states	$\rho_1 \otimes \dots \otimes \rho_n$	$ \psi_1\rangle \otimes \dots \otimes \psi_n\rangle$

EXAMPLE:

A state is represented by the following density matrix.

$$\rho = \begin{pmatrix} \frac{2}{3} & \frac{1}{6} - \frac{i}{3} \\ \frac{1}{6} + \frac{i}{3} & \frac{1}{3} \end{pmatrix}$$

Show that this is a mixed state.

calculate the Bloch vector and its magnitude.

$$\text{tr}(\rho) \rightarrow 1 \quad \text{tr}(\rho^2) \rightarrow \frac{5}{6}$$

$$R_x = \text{tr}(\rho\sigma_x) \rightarrow \frac{1}{3} \quad R_y = \text{tr}(\rho\sigma_y) \rightarrow \frac{2}{3}$$

$$R_z = \text{tr}(\rho\sigma_z) \rightarrow \frac{1}{3}$$

$$R = \sqrt{R_x^2 + R_y^2 + R_z^2} = 0.816$$

Distance metrics :

Trace distance

We begin by defining the *trace distance* between quantum states ρ and σ ,

$$D(\rho, \sigma) \equiv \frac{1}{2} \text{tr}|\rho - \sigma|.$$

where as per usual we define $|A| \equiv \sqrt{A^\dagger A}$ to be the positive square root of $A^\dagger A$.

EXAMPLE:

Find the trace distance between the states

$$\rho = \begin{pmatrix} \frac{5}{8} & \frac{i}{4} \\ -\frac{i}{4} & \frac{3}{8} \end{pmatrix}, \quad \sigma = \begin{pmatrix} \frac{2}{5} & \frac{-i}{8} \\ \frac{i}{8} & \frac{3}{5} \end{pmatrix}$$

$$\rho - \sigma = \begin{pmatrix} \frac{5}{8} & \frac{i}{4} \\ -\frac{i}{4} & \frac{3}{8} \end{pmatrix} - \begin{pmatrix} \frac{2}{5} & \frac{-i}{8} \\ \frac{i}{8} & \frac{3}{5} \end{pmatrix} = \begin{pmatrix} \frac{9}{40} & \frac{i3}{8} \\ -\frac{i3}{8} & \frac{-9}{40} \end{pmatrix} \quad (\rho - \sigma)^\dagger (\rho - \sigma) = \begin{pmatrix} \frac{9}{40} & \frac{i3}{8} \\ -\frac{i3}{8} & \frac{-9}{40} \end{pmatrix} \begin{pmatrix} \frac{9}{40} & \frac{i3}{8} \\ -\frac{i3}{8} & \frac{-9}{40} \end{pmatrix} = \begin{pmatrix} \frac{153}{800} & 0 \\ 0 & \frac{153}{800} \end{pmatrix}$$

Next we find

$$|\rho - \sigma| = \sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)} = \sqrt{\begin{pmatrix} \frac{153}{800} & 0 \\ 0 & \frac{153}{800} \end{pmatrix}} = \frac{1}{20} \begin{pmatrix} 3\sqrt{\frac{17}{2}} & 0 \\ 0 & 3\sqrt{\frac{17}{2}} \end{pmatrix} \quad \delta(\rho, \sigma) = \frac{1}{2} \left(\frac{1}{20} \right) (2) \left(3\sqrt{\frac{17}{2}} \right) \approx 0.437$$

Fidelity

A second measure of distance between quantum states is the *fidelity*.

The fidelity of states ρ and σ is defined to be

$$F(\rho, \sigma) \equiv \text{tr} \sqrt{\rho^{1/2} \sigma \rho^{1/2}}.$$

Reduced density operator:

If we have a composite system composed by individual subsystems, $\rho_1 \otimes \dots \otimes \rho_n$ we may need to know the quantum state of the individual subsystems. Then we have to find the reduced density operators corresponding to those subsystems.

Suppose we have physical systems A and B , whose state is described by a density operator ρ^{AB} .

The reduced density operator for system A is defined by where tr_B is a map of operators known as

$$\rho^A \equiv \text{tr}_B(\rho^{AB}),$$

the partial trace over system B .

Likewise,

$$\rho^B = \text{tr}_A(\rho^{AB}).$$

The partial trace is defined by

$$\text{tr}_B (|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) \equiv |a_1\rangle\langle a_2| \text{tr}(|b_1\rangle\langle b_2|),$$

where $|a_1\rangle$ and $|a_2\rangle$ are any two vectors in the state space of A , and $|b_1\rangle$ and $|b_2\rangle$ are any two vectors in the state space of B .

Make it clear:

$$\rho^A = \text{tr}_B(\rho^{AB}), \quad \text{Remember that } \text{Tr}(A) = \sum A_{ii} = \langle 0|A|0\rangle + \langle 1|A|1\rangle$$

$$\rho^A = \text{tr}_B(\rho^{AB}) = (\mathbb{1} \otimes \langle 0|) (\rho^{AB}) (\mathbb{1} \otimes |0\rangle) + (\mathbb{1} \otimes \langle 1|) (\rho^{AB}) (\mathbb{1} \otimes |1\rangle)$$

$$\rho^B = \text{tr}_A(\rho^{AB}) = (\langle 0| \otimes \mathbb{1}) (\rho^{AB}) (|0\rangle \otimes \mathbb{1}) + (\langle 1| \otimes \mathbb{1}) (\rho^{AB}) (|1\rangle \otimes \mathbb{1})$$

One can also check the dimensions by matrix multiplications.

Take, $(\langle 0| \otimes \mathbb{1}) (\rho^{AB}) (|0\rangle \otimes \mathbb{1})$;

$$\left(\begin{bmatrix} 1 & 0 \end{bmatrix}_{1 \times 2} \otimes \begin{bmatrix} 1 & 1 \end{bmatrix}_{2 \times 2} \right) [\rho^{AB}]_{4 \times 4} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}_{2 \times 1} \otimes \begin{bmatrix} 1 & 1 \end{bmatrix}_{2 \times 2} \right)$$

$$= [\langle 0| \otimes \mathbb{1}]_{2 \times 4} [\rho^{AB}]_{4 \times 4} [|0\rangle \otimes \mathbb{1}]_{4 \times 2}$$

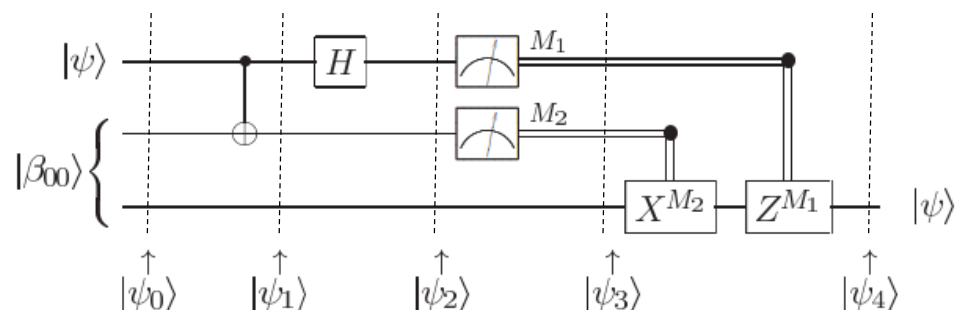
$$= [(\langle 0| \otimes \mathbb{1}) \rho^{AB}]_{2 \times 4} [|0\rangle \otimes \mathbb{1}]_{4 \times 2}$$

Therefore,

$$[\rho^B]_{2 \times 2} = [(\langle 0| \otimes \mathbb{1}) \rho^{AB} (|0\rangle \otimes \mathbb{1})]_{2 \times 2} + [(\langle 1| \otimes \mathbb{1}) \rho^{AB} (|1\rangle \otimes \mathbb{1})]_{2 \times 2}$$

EXAMPLE: *Teleportation Revisited*

Consider the quantum teleportation circuit and assume that instead of applying the standard protocol, one tries to find Bob's qubit by taking the partial trace over Alice's qubits, what is the result?



Recall that immediately before Alice makes her measurement the quantum state of the three qubits is

$$|\psi_2\rangle = \frac{1}{2} \left[|00\rangle (\alpha|0\rangle + \beta|1\rangle) + |01\rangle (\alpha|1\rangle + \beta|0\rangle) + |10\rangle (\alpha|0\rangle - \beta|1\rangle) + |11\rangle (\alpha|1\rangle - \beta|0\rangle) \right].$$

Measuring in Alice's computational basis, the state of the system after the measurement is:

$$\begin{array}{ll} |00\rangle [\alpha|0\rangle + \beta|1\rangle] & \text{with probability } \frac{1}{4} \\ |01\rangle [\alpha|1\rangle + \beta|0\rangle] & \text{with probability } \frac{1}{4} \\ |10\rangle [\alpha|0\rangle - \beta|1\rangle] & \text{with probability } \frac{1}{4} \\ |11\rangle [\alpha|1\rangle - \beta|0\rangle] & \text{with probability } \frac{1}{4}. \end{array}$$

The density operator of the system is thus

$$\rho = |\psi_2\rangle\langle\psi_2| = \frac{1}{4} \left[|00\rangle\langle 00|(\alpha|0\rangle + \beta|1\rangle)(\alpha^*\langle 0| + \beta^*\langle 1|) + |01\rangle\langle 01|(\alpha|1\rangle + \beta|0\rangle)(\alpha^*\langle 1| + \beta^*\langle 0|) \right. \\ \left. + |10\rangle\langle 10|(\alpha|0\rangle - \beta|1\rangle)(\alpha^*\langle 0| - \beta^*\langle 1|) + |11\rangle\langle 11|(\alpha|1\rangle - \beta|0\rangle)(\alpha^*\langle 1| - \beta^*\langle 0|) \right].$$

$$\rho^B = \frac{1}{4} \left[(\alpha|0\rangle + \beta|1\rangle)(\alpha^*\langle 0| + \beta^*\langle 1|) + (\alpha|1\rangle + \beta|0\rangle)(\alpha^*\langle 1| + \beta^*\langle 0|) + (\alpha|0\rangle - \beta|1\rangle)(\alpha^*\langle 0| - \beta^*\langle 1|) + (\alpha|1\rangle - \beta|0\rangle)(\alpha^*\langle 1| - \beta^*\langle 0|) \right] \\ = \frac{2(|\alpha|^2 + |\beta|^2)|0\rangle\langle 0| + 2(|\alpha|^2 + |\beta|^2)|1\rangle\langle 1|}{4}$$

where we have used the completeness relation in the last line. Thus, the state of Bob's system *after* Alice has performed the measurement but *before* Bob has learned the measurement result is $I/2$. This state has no dependence upon the state $|\psi\rangle$ being teleported, and thus any measurements performed by Bob will contain no information about $|\psi\rangle$, thus preventing Alice from using teleportation to transmit information to Bob faster than light.